

38.5 Summary and Notes

38.5.1 Summary

Idea.

1. A **variance specification** (Definition 38.1.1) gives the mean and the variance of each response and nothing else: $EY_i = h(\mathbf{x}_{(i)}^\top \boldsymbol{\beta})$ and $\text{Var}(Y_i) = \phi V(\mu_i)/w_i$, with ϕ free even for a variance function borrowed from a family that fixed it. No distribution need exist with these moments, and for ungrouped binary data none does with $\phi \neq 1$. Overdispersion of a group proportion has two shapes, a constant multiple of the binomial variance and the factor $1 + \rho(m-1)$ of exchangeable trials (Lemma 38.1.1), agreeing at one group size only.
2. A variance function fixes the estimating equations, not the fit: the NB1 model shares the quasi-Poisson variance function and gives a different $\hat{\boldsymbol{\beta}}$ (Example (Four analyses of the strike data)).
3. The **quasi-likelihood** $Q(\mu; y) = w\phi^{-1} \int_y^\mu (y-t)V(t)^{-1} dt$ of Definition 38.2.1 has the score, the two Bartlett identities and the saturated maximum of a log-likelihood, and equals the log-likelihood minus its saturated value for an exponential dispersion family (Proposition 38.2.1); its -2ϕ multiple is the **quasi-deviance**.
4. The **quasi-score** is the score Eq. 34.3.1, so the estimate is the usual iteratively reweighted least squares fit (Theorem 38.2.2(a)), unbiased as soon as the mean model is right whatever the variances; the identity between its variance and its sensitivity needs the variance model too (Theorem 38.2.2(b)–(e)).
5. Among **linear estimating functions** $\mathbf{C}(\boldsymbol{\beta})(\mathbf{y} - \boldsymbol{\mu})$ the quasi-score maximizes the Godambe information, and nothing else does except its nonsingular linear transformations (Proposition 38.2.3). The proof is Gauss–Markov with an orthogonal projection in place of the completion of the square.
6. Under (Q1)–(Q5), $\hat{\boldsymbol{\beta}}$ is asymptotically normal with the **sandwich** covariance $\mathbf{A}_n^{-1} \mathbf{B}_n \mathbf{A}_n^{-1}$, whose middle is estimated from the residuals (Theorem 38.3.1); a correct variance specification makes $\mathbf{B}_n = \mathbf{A}_n$ and collapses it to the model-based $\phi(\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1}$, then valid and more efficient.
7. The dispersion is estimated by $X^2/(n-p)$, of expectation exactly ϕ at the true mean, while the deviance version is not in general consistent (Proposition 38.3.2(a),(b)). Nested models are compared by the quasi- F statistic Eq. 38.3.3, which needs a correct variance function; the Akaike criterion has no analogue here beyond a convention (Proposition 38.3.2(c),(d)).
8. Overdispersion comes from unobserved heterogeneity, clustering and contagion, each computable exactly (Proposition 38.4.1(a)–(c)). Ignoring a dispersion ϕ_0 turns a nominal 5% test of one coefficient into one of size $\Pr\{\chi^2(1) > 3.841/\phi_0\}$, and of six coefficients into something far worse (Eq. 38.4.2). The repairs assume different things — a mean and a variance shape, a full distribution, a mixing law — and the last changes the estimand.

38.5.2 Notes and sources

Wedderburn’s paper. Quasi-likelihood is due to Wedderburn (1974), written two years after the generalized linear model itself (Nelder and Wedderburn 1972) and in the same spirit: notice which part of the structure the algorithm uses, and keep only that. The definition Eq. 38.2.1, the two Bartlett identities and the table of quasi-likelihoods are all in that paper, as is the name. McCullagh and Nelder (1989, chapter 9) is the standard extended treatment; their chapter 10 develops the joint model for the mean and the dispersion, which this chapter does not cover. Multiplying Poisson or binomial standard errors by $\sqrt{X^2/(n-p)}$ predates the theory — Finney (1947) used it for probit analysis — and Wedderburn said what it estimates.

Estimating functions and optimality. The general theory of unbiased estimating functions is Godambe (1960), who showed the likelihood score optimal among them in the sense used in Proposition 38.2.3; Godambe and Heyde (1987) survey the subject. The projection proof given here makes the analogy exact: “optimal among linear estimating functions” stands to “optimal among all” as “best linear unbiased” stands to “minimum variance unbiased” in Theorem 7.2.1. McCullagh (1983) proved the asymptotics under conditions close to (Q1)–(Q5).

The sandwich. Its history is traced in Section 21.4: Eicker (1963), Huber (1967) and White (1980) in three literatures at once. The generalized-linear-model version, with the working weights in the bread, is in Royall (1986), who argued for it as the default; Liang and Zeger (1986) made it the foundation of generalized estimating equations. Its small-sample bias is analysed by Kauermann and Carroll (2001), and the leverage corrections of Example (Coverage when the variance function is right and when it is wrong) are the HC2 and HC3 of MacKinnon and White (1985).

Binomial overdispersion. The exchangeable-trials calculation Eq. 38.1.3 is the design effect of cluster sampling (Kish 1965), which Chapter 33 met as $1 + (m-1)\rho$. The beta-binomial distribution is due to Skellam (1948), and Williams (1982) proposed fitting the variance function $\{1 + \rho(m_i - 1)\}\mu(1 - \mu)/m_i$ by alternating a quasi-likelihood step with a moment equation setting X^2 equal to its degrees of freedom.

Which device to use. Ver Hoef and Boveng (2007) is a clear applied statement of the quasi-Poisson against negative binomial question, showing that the choice is between the NB1 and NB2 variance functions and matters most where the fitted means are most spread out — what Example (Four analyses of the strike data) shows in miniature. Cameron and Trivedi (2013) covers the quasi-likelihood, negative binomial and sandwich routes together. The “QAIC” convention of Proposition 38.3.2(d) is from Lebreton, Burnham, Clobert and Anderson (1992), widely used in ecology and resting on no theorem.

The data. The strike durations of Example (Strike durations) are the subset of Kennan’s (1985) contract-strike series distributed with statsmodels, from Bureau of Labor Statistics records. Kennan fitted duration models of a different kind; the Poisson mean model here illustrates the methods of this chapter and is not an account of strike durations.

What comes next. Chapter 39 returns to models with likelihoods, taking up model building, diagnostics and the Bayesian treatment. Chapter 40 extends the estimating-equation idea to correlated responses, where the working variance becomes a working *correlation matrix*. Chapter 41 asks what happens when some responses are not observed at all.

38.5.3 References

- Cameron, A. Colin and Trivedi, Pravin K. (2013). *Regression Analysis of Count Data*. 2nd edition. Cambridge: Cambridge University Press.
- Eicker, Friedhelm (1963). Asymptotic Normality and Consistency of the Least Squares Estimators for Families of Linear Regressions. *The Annals of Mathematical Statistics* 34(2), 447–456.
- Fahrmeir, Ludwig and Kaufmann, Heinz (1985). Consistency and Asymptotic Normality of the Maximum Likelihood Estimator in Generalized Linear Models. *The Annals of Statistics* 13(1), 342–368.
- Finney, David J. (1947). *Probit Analysis: A Statistical Treatment of the Sigmoid Response Curve*. Cambridge: Cambridge University Press.
- Godambe, V. P. (1960). An Optimum Property of Regular Maximum Likelihood Estimation. *The Annals of Mathematical Statistics* 31(4), 1208–1211.
- Godambe, V. P. and Heyde, C. C. (1987). Quasi-Likelihood and Optimal Estimation. *International Statistical Review* 55(3), 231–244.
- Huber, Peter J. (1967). The Behavior of Maximum Likelihood Estimates under Non-standard Conditions. In *Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability*, volume 1, 221–233. Berkeley: University of California Press.
- Kauermann, Göran and Carroll, Raymond J. (2001). A Note on the Efficiency of Sandwich Covariance Matrix Estimation. *Journal of the American Statistical Association* 96(456), 1387–1396.
- Kennan, John (1985). The Duration of Contract Strikes in US Manufacturing. *Journal of Econometrics* 28(1), 5–28.
- Kish, Leslie (1965). *Survey Sampling*. New York: Wiley.
- Lebreton, Jean-Dominique, Burnham, Kenneth P., Clobert, Jean and Anderson, David R. (1992). Modeling Survival and Testing Biological Hypotheses Using Marked Animals: A Unified Approach with Case Studies. *Ecological Monographs* 62(1), 67–118.
- Liang, Kung-Yee and Zeger, Scott L. (1986). Longitudinal Data Analysis Using Generalized Linear Models. *Biometrika* 73(1), 13–22.
- MacKinnon, James G. and White, Halbert (1985). Some Heteroskedasticity-Consistent Covariance Matrix Estimators with Improved Finite Sample Properties. *Journal of Econometrics* 29(3), 305–325.
- McCullagh, Peter (1983). Quasi-Likelihood Functions. *The Annals of Statistics* 11(1), 59–67.
- McCullagh, Peter and Nelder, John A. (1989). *Generalized Linear Models*. 2nd edition. London: Chapman and Hall.
- Nelder, John A. and Wedderburn, Robert W. M. (1972). Generalized Linear Models. *Journal of the Royal Statistical Society, Series A* 135(3), 370–384.
- Royall, Richard M. (1986). Model Robust Confidence Intervals Using Maximum Likelihood Estimators. *International Statistical Review* 54(2), 221–226.
- Skellam, J. G. (1948). A Probability Distribution Derived from the Binomial Distribution by Regarding the Probability of Success as Variable between the Sets of Trials. *Journal of the Royal Statistical Society, Series B* 10(2), 257–261.
- Ver Hoef, Jay M. and Boveng, Peter L. (2007). Quasi-Poisson vs. Negative Binomial Regression: How Should We Model Overdispersed Count Data? *Ecology* 88(11), 2766–2772.
- Wedderburn, Robert W. M. (1974). Quasi-Likelihood Functions, Generalized Linear Models, and the Gauss–Newton Method. *Biometrika* 61(3), 439–447.
- White, Halbert (1980). A Heteroskedasticity-Consistent Covariance Matrix Estimator and a Direct Test for Heteroskedasticity. *Econometrica* 48(4), 817–838.

-
- Williams, D. A. (1982). Extra-Binomial Variation in Logistic Linear Models. *Journal of the Royal Statistical Society, Series C (Applied Statistics)* 31(2), 144–148.